

M003 Week 1 — English Worksheet

Topic: First Steps with Coordinates (x, y) · **Course:** 3D Coordinates · **Time:** about 45 minutes

This week you draw on a flat **wall**. Every spot on the wall gets a name made of **two numbers: (x, y)**. x tells you how far **across**, y tells you how far **up**. Counting starts at **0**.

1 · Predict 🧠

Read each set of steps. Before you imagine it happening, write what you think you will see.

```
place gold block at (3, 0)
place gold block at (3, 5)
```

The two gold blocks share the same x. Are they close together or far apart? Which number changed?


```
place red block at (0, 0)
place red block at (4, 0)
place red block at (0, 4)
```

Three red blocks. Which one sits at the corner where counting starts? Which one is higher up the wall?


```
set y to 2
place gold block at (0, y)
set y to 6
place gold block at (0, y)
```

The variable y changes from 2 to 6. Where does each gold block land? Do they share the same x ?

2 · Spot the Bug 🐛

Each block of code below was meant to do something, but it is broken. Read what the code is **supposed** to do, then rewrite it so it works. After that, explain why the original was wrong and why your fix works.

Bug A — The block should land at **(2, 5)** — 2 across on x, 5 up on y. Right now it lands somewhere else.

```
place gold block at (5, 2)
```

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug B — The block should sit at the **bottom corner** of the wall. The coder forgot where counting starts.

```
place gold block at (1, 1)
```

Hint: counting starts at **0**, not 1.

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug C — The block should sit **5 up** from the ground at $x = 0$ — that is **(0, 5)**. Right now it goes the wrong way and drops below the wall.

```
place gold block at (0, -5)
```

Write the fixed code:

Why was it wrong? Why does your fix work?

3 · Tell Me What You Built 📷

Now switch to your homework world. Place a gold block at **(0, 0)** so you can see where counting starts. Then pick three more coordinates, place a different block at each one, and write down the coordinates you used. When you finish, come back here.

Send a photo or video of your blocks, then explain what you did. Use these sentence starters — write 4 to 6 sentences total.

First, I ...

The block at **(0, 0)** shows ...

I placed a block at $(\dots, \dots)$ because ...

The y number means ...

One tricky moment was when ...

If I had more time, I would ...

4 · Record Your Walkthrough

Now take a video on your phone (or a parent's phone) while you show your blocks on the wall. Talk through it like you are teaching someone who has never seen coordinates. Try to use these words: **coordinate, position, x, y, zero.**

1. Show the gold block at (0, 0) and say why it is there.
2. Point at one of your blocks and say its coordinate out loud.
3. Pick two blocks and say which one is **higher** and how you know from the y number.
4. Say in your own words what **x** and **y** mean.

Write what you will say in your video. Use the space below to plan it before you record — you can read from it while filming.

Submit 

Send this worksheet + your walkthrough video to teacher on KakaoTalk.